

Digital Implementation of Enhanced Phase-Locked Loop for Electrical Grid Synchronization

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Abstract— This paper describes the implementation of an enhanced phase-locked loop and its results as grid synchronization for grid-tied power converters. The proposed structure is compared to a simpler phase-locked loop based on synchronous reference frame. A C-based dedicated library called **renolib** is used for their implementation in a digital signal processor. Both phase-locked loop structures are tested in two different scenarios: a digitally generated three-phase voltage system and a real three-phase power system, fed by the low voltage electrical distribution network.

I. INTRODUCTION

The use of grid-tied power converters includes not only industrial devices, such as uninterrupted power supplies, power rectifiers and battery chargers, but also its several applications in energy generation (mainly renewables power sources, like photovoltaic and wind [1]), storage [2] and transmission and distribution [3].

The control of such type of power converters requires the estimation of instantaneous magnitude and phase of the grid voltage. A three phase Phase-Locked Loop (PLL) is used to represent grid voltage in different reference frames (such as rotating and stationary) and in consequence to obtain the desired magnitudes. This process is commonly known as grid synchronization.

The PLL is a critical system for the control of grid-connected converters. Particularly, unbalanced and distorted electrical grids can deteriorate the PLL performance in steady state. Besides, dynamic response of PLL to electrical grid faults must be fast and stable to keep the converter synchronized to the grid during voltage disturbances such as sags and swells. As a result, it is highly recommended that grid-tied converters include a robust type of PLL to maintain the grid synchronization process in different operating conditions.

Many versions of PLL structures are devoted to improving its steady and dynamic response. [4] presents a self-synchronization mechanism based on the converter interactions with grid impedance and power flow directions. A robust version of a PLL for power quality is presented in [5]. Optimized versions and efficient computational implementation can be found in [6], [7] and [8], whereas an hybrid structure that includes zero cross detection (ZCD) is presented in [9].

This paper focuses on the implementation of an enhanced PLL structure called Enhanced PLL II (EPLLII) [10]. The proposed PLL will be used for grid synchronization of power converters at low voltage distribution network. Its steady state operation and its dynamic response is compared to a second implementation consisting of a synchronous

reference frame (SRF) PLL. A universal power converter controller is used as a hardware platform [11] for the digital implementation of both PLL algorithms.

II. THREE-PHASE SRF-PLL

A. General aspects

PLL are used to determine the fundamental positive sequence component of grid voltage. It includes the instantaneous estimation of magnitude, frequency and phase. In a power system, all these variables may be subject of different type of disturbances and, therefore, they are continuously changing. PLL systems must be fast enough to rapidly lock the grid voltage in case of such disturbances. Generator, lines and load disconnections and different types of grid faults are among the most common contingencies in electrical power systems.

Voltage unbalance and harmonics, mainly caused by nonlinear loads, may also deteriorate the steady state operation of PLL systems in highly distorted electrical grids.

Three-phase PLL mechanisms estimate both magnitude, frequency and phase of the corresponding spatial vectors by measuring the grid voltages (abc reference frame). Since space vector representation is usually carried out in stationary $\alpha\beta$ or rotating dq reference frames, these PLL algorithms include Clarke and Park transformations to convert the measured voltages and currents to the chosen reference frame.

It is important to emphasize the fact that three-phase PLL must not be understood as three single-phase PLL, as the three phases of a power system do not work autonomously but does work in a coordinated way [1].

B. SRF-PLL operation in ideal case

The operating principle for SRF-PLL structure is shown in Fig. 1. It assumes an ideal case where the grid voltage is symmetrical and balanced, and no voltage harmonics are present. Its space vector representation is shown as V_G . It rotates at $2\pi f$, where f is the electrical frequency. A rotating dq reference frame is internally generated by the PLL. It is shown in Fig. 1 as axes d and q . They are orthogonal to each other, where q leads d . The speed of the rotating reference frame can be controlled by the PLL, and it is noted as ω_{PLL} . The angular position of the reference frame is given by angle ϕ , which is equivalent to the time integration of ω_{PLL} .

In energy applications, the PLL will be phase-locked to the grid voltage when the phase difference between spatial vector V_G and the d axis is set to zero. Thus, the orthogonal projection of space vector V_G over the q axis, named v_{Gq} , is zero.

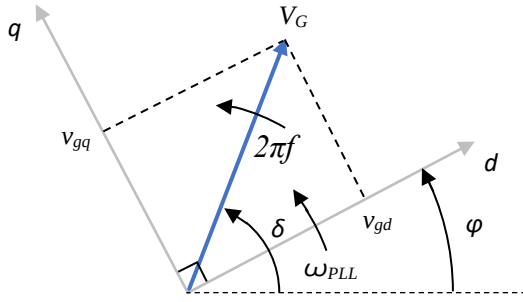


Fig. 1. Vector representation of grid voltage and internally generated dq reference frame

An automatic mechanism for grid synchronization could force v_{Gq} to zero by means of a closed loop control. This goal can be achieved by means of two cascaded integrators, as shown in Fig. 2. The first integrator processes the quadrature component of the grid voltage, v_{Gq} , accelerating the synchronous reference frame by increasing ω_{PLL} . The synchronous reference frame will be accelerated when v_{Gq} is greater than zero, whereas it will be decelerated when v_{Gq} becomes lower than zero.

Phase angle of the PLL can be estimated by the time integration of ω_{PLL} , which is accomplished by the second integrator.

The steady state condition for a phase-locked condition is shown in Fig. 2. In this case, v_{Gq} is set to zero by the cascaded integrators and the synchronous reference frame is tracked to the grid voltage vector, given by V_G .

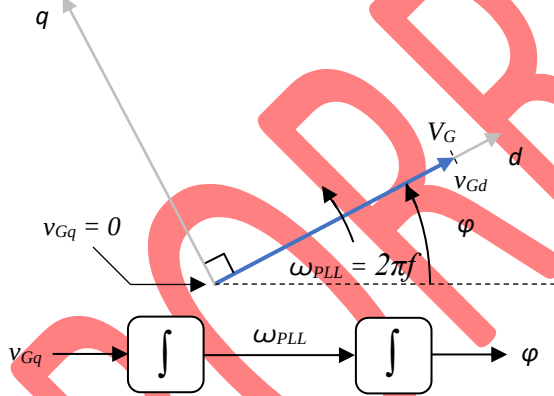


Fig. 2. Phase-locked operation of PLL and space vector of grid voltage.

As the PLL is synchronized to the grid, instantaneous magnitude of the grid voltage can be measured by v_{Gd} magnitude (being the orthogonal projection of V_G on the d axis), whereas its frequency and phase angle can be estimated by the rotating speed and phase angle of the synchronous reference frame (variables ω_{PLL} and ϕ , respectively). Fig. 3 shows the evolution of these variables under non-distorted conditions (blue traces). The shown angle is the difference between the real phase of the electrical system and the phase of the PLL, given by ϕ . Since the PLL is synchronized to the grid, both angles are equal, and their resulting difference is zero. In steady state conditions, angle ϕ is a sawtooth waveform between 0 a 2π radians and its frequency is given by the electrical frequency of the grid.

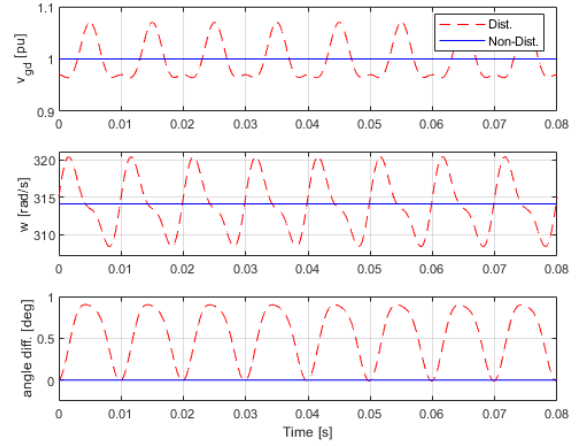


Fig. 3. Estimated fundamental positive sequence variables by SRF-PLL in distorted (red) and non-distorted (blue) grid conditions.

C. SRF-PLL operation in distorted grids

The presence of harmonics and unbalanced operating conditions affects the steady state performance of the SRF-PLL. In those cases, the space vector representation of grid voltage is no longer a single-frequency vector but a sum of space vectors with different rotating speeds. Each of these speeds will depend on the harmonic number of the component and its type of sequence (positive, negative or zero). Fig. 4 shows an example of the resulting space vector for an arbitrary case where the grid is unbalanced (presence of fundamental negative sequence, noted as V_{G1-}) and distorted (presence of positive sequence fifth harmonic, noted as V_{G5+}). Its fundamental positive sequence is identified by V_{G1+} .

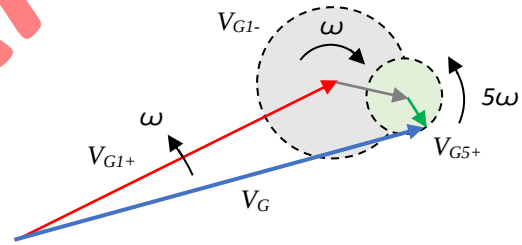


Fig. 4. Space vector representation for grid voltage with voltage unbalance (fundamental neg. seq.) and presence of pos. seq. fifth harmonic.

Fig. 3 plots the estimated positive sequence variables by a SRF-PLL when both voltage unbalance and harmonics are present (red traces). The example under assessment considers a 5% voltage unbalance (fundamental negative sequence) and a 2% harmonic content (positive sequence fifth harmonic).

Both voltage unbalance and harmonics affects the estimated magnitude of the fundamental positive sequence as two different harmonic components are now superimposed to its dc value. A fourth-harmonic component is caused by the interaction of the synchronous reference frame with the fifth-harmonic positive sequence component, whereas a second harmonic is produced by the interaction with the fundamental negative sequence.

Since positive sequence k -harmonics rotate in the same direction as the synchronous reference frame, their effect in estimated variables of the PLL are pulsations whose frequency are $(k - 1)$ times the nominal frequency. On the

other hand, negative sequence k -harmonics rotate in the opposite direction, leading to the appearance of $(k + 1)$ harmonics in the estimated variables of the PLL. Zero sequence components do not produce any type of deviations nor pulsations in the estimated variables of the SRF-PLL.

The presence of voltage unbalance and harmonics not only affects the estimated magnitude as the estimated angle difference and frequency have also superimposed oscillations. The impact over these variables is attenuated by the low-pass dynamic response of the cascaded integrators.

As the PLL is a critical subsystem for grid-tied converters, unproper operation or malfunction on this sort of equipment may occur in highly distorted ac grids.

D. Block diagram

A continuous time domain block diagram for SRF-PLL is shown in Fig. 5. The measured voltage, v_G , in abc coordinates is transformed to $\alpha\beta$ and dq reference frames by means of Clarke and Park transforms, respectively.

A proportional plus integral controller (PI) is used to accelerate or decelerate the rotating reference frame of the SRF-PLL. This controller replaces the first integrator shown in Fig. 2, introducing a proportional gain to the error to improve the dynamic performance of the overall system.

The phase of the synchronous reference frame, φ , is calculated as the time integration of the sum of two components: the nominal system frequency, ω_N , and the PI controller output, $\Delta\omega$.

The PI controller will try to force v_{Gq} to zero in steady state. As a result, the internally generated rotating reference frame for the PLL will be locked to the spatial vector of grid voltage.

Transients and external disturbances may affect the synchronization process of the SRF-PLL. In these cases, the dynamic response of the overall system will be determined by the proportional and integral gains of the controller and the phase integrator.

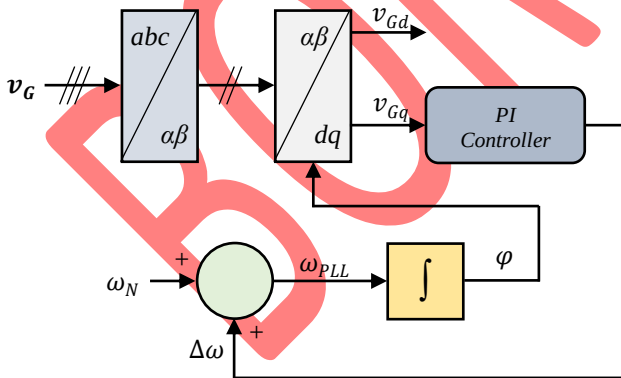


Fig. 5. Continuous time domain block diagram for SRF-PLL.

The block diagram of Fig. 5 is used for the digital implementation of SRF-PLL, as it will be presented in section V.

III. THREE-PHASE ENHANCED-PLL

In addition to the instantaneous estimation of phase, frequency and magnitude of fundamental positive sequence, the EPLLII allows the use of multiple cores to estimate the amplitude and phase of different sequences and harmonics present in the measured voltage.

A. Structure

Fig. 6 shows a generic representation for the EPLLII structure. The number and type of cores is defined by the user, although the inclusion of fundamental positive sequence core is mandatory. Three-phase signals are marked with '///' to distinguish them from scalar variables.

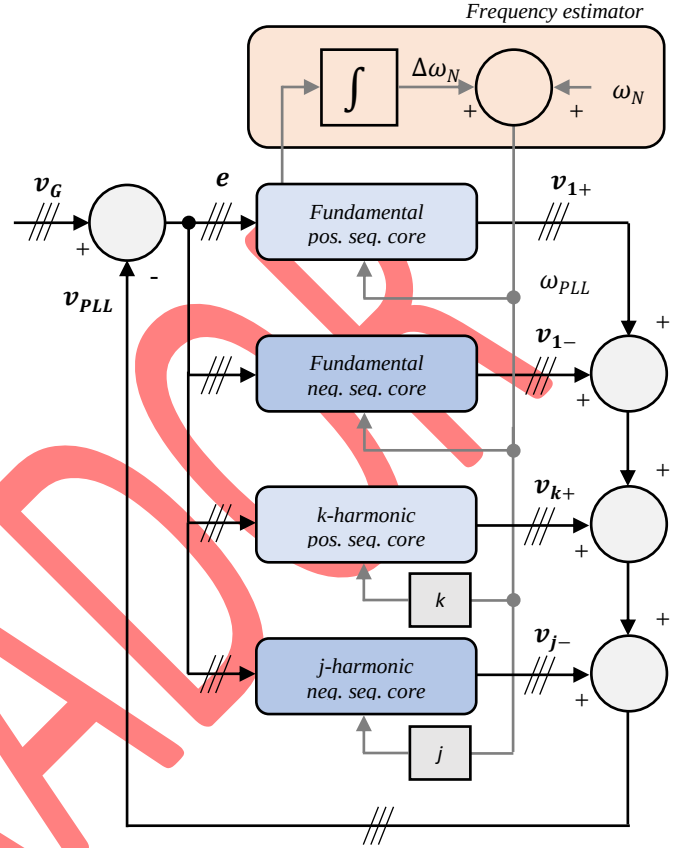


Fig. 6. A generic representation for multiple cores in EPLLII.

Each core allows the EPLLII to estimate a particular sequence and harmonic of the measured voltage. In the example of Fig. 6, the type of cores has been selected to detect: i) fundamental pos. seq. (v_{1+}), ii) fundamental neg. seq. (v_{1-}), iii) k -harmonic pos. seq. (v_{k+}) and, iv) j -harmonic neg. seq. (v_{j-}).

Each core output is a three-phase single-frequency variable set, whose frequency and type of component will depend on the selected type of core. Its output magnitude will depend on the amplitude for that specific sequence and harmonic in the measured grid voltage.

The sum of all cores, named as v_{PLL} , is a digital three-phase representation of the instantaneous value of the grid voltage. All three-phase variables in this PLL structure are referenced to the abc reference frame.

The individual detection of multiple sequences and harmonics by each core allows the EPLLII to cancel their effects in the estimation of the fundamental positive sequence, v_{1+} . It can be considered as an important strength of the EPLLII compared to SRF-PLL, as in the later the presence of voltage unbalance and harmonics may deteriorate its steady state operation (as shown in Fig. 3).

As the electrical grid frequency may slightly differ from its nominal value, a frequency estimator core is used. This block estimates the fundamental frequency of the electrical grid based on its fundamental positive sequence component,

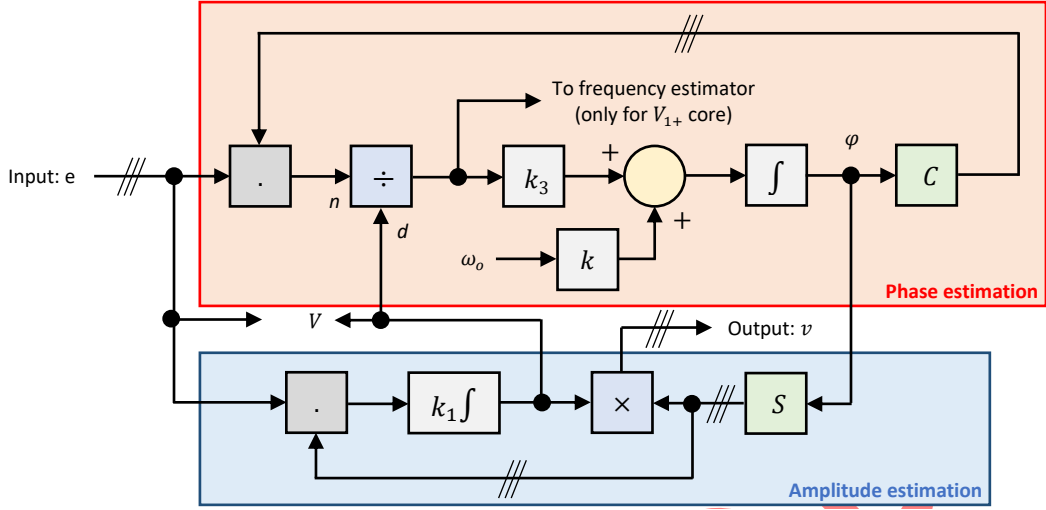


Fig. 7. Generic sequence core estimator for EPLLII.

since its amplitude is expected to be largely superior to all other sequences and harmonics.

B. Modularity

The type of structure for EPLLII is clearly modular. Additional cores can be introduced to detect a larger number of sequences or harmonics.

The limit of sequences or harmonics to be estimated is exclusively limited by the capacity of the CPU to execute all cores in an instruction cycle. The designer can also rely on some features of electrical power systems for choosing the most convenient cores for its application, such as:

- Positive and negative sequence components can be found on odd harmonics that are not multiples of 3 ($k = 5, 7, 11, 13 \dots$).
- Zero sequence components can only be found in four-wire configurations (three phase plus neutral) and its harmonics are multiples of 3.
- Even harmonics are not usually present at electrical power systems and can be omitted.

C. Estimation cores

A generic representation for a single core is shown in Fig. 7. The structure is common for positive, negative and zero sequence cores. They only differ in the definition of functions $C(\varphi)$ and $S(\varphi)$, as presented in Table 1.

Constant k is defined in each core to specify the harmonic to be detected, being an integer variable. It is user-selectable according to the technical needs. k can be set to unity to estimate fundamental components.

The structure of the generic core consists of two main parts: phase and magnitude estimation. Both parts are highlighted in Fig. 5 in red and blue colour, respectively.

The input error, named as variable e , is a three-phase signal that it is defined as the difference between the measured voltage and v_{PLL} . It is used by both magnitude and phase estimation parts of all cores, as shown in Fig. 6.

Phase estimation of EPLLII uses a phase detector given by a multiplier block. The dynamic response of this control loop can be adjusted by means of gain k_3 . The output of this proportional controller plus the frequency estimation term, $k\omega_o$, will determine the instantaneous frequency of the

sequence under assessment. Its time integration will result in its instantaneous phase, φ .

Seq.	$C(\varphi)$ block	$S(\varphi)$ block
+	$\begin{bmatrix} \cos(\varphi) \\ \cos(\varphi - 2\pi/3) \\ \cos(\varphi + 2\pi/3) \end{bmatrix}$	$\begin{bmatrix} \sin(\varphi) \\ \sin(\varphi - 2\pi/3) \\ \sin(\varphi + 2\pi/3) \end{bmatrix}$
-	$\begin{bmatrix} \cos(-\varphi) \\ \cos(-\varphi - 2\pi/3) \\ \cos(-\varphi + 2\pi/3) \end{bmatrix}$	$(-1) \cdot \begin{bmatrix} \sin(-\varphi) \\ \sin(-\varphi - 2\pi/3) \\ \sin(-\varphi + 2\pi/3) \end{bmatrix}$
0	$\begin{bmatrix} \cos(\varphi) \\ \cos(\varphi) \\ \cos(\varphi) \end{bmatrix}$	$\begin{bmatrix} \sin(\varphi) \\ \sin(\varphi) \\ \sin(\varphi) \end{bmatrix}$

Table 1. $C(\varphi)$ and $S(\varphi)$ function definition for each type of sequence.

As the amplitude of the measured voltages acts as a gain in the open loop transfer function of the EPLL-II, an improvement in the dynamic response of the phase estimation loop is achieved by means of an amplitude normalization, given by a divider block whose denominator is the calculated amplitude of the sequence, given by V . This functionality allows the EPLLII to maintain its dynamic response performance during large deviations of the amplitude of the measured voltage. Amplitude normalization can be implemented in other type of PLL structures, such as SRF-PLL.

Magnitude estimation is a closed-loop structure whose dynamic response may be adjusted through gain k_1 . Usually, gains k_1 and k_3 are the same for all cores.

The output of each core is a three-phase signal. All core outputs are externally summed, resulting variable v_{PLL} . It is used in the calculation of the error, given by e , being subtracted from the measured grid voltages.

IV. HARDWARE IMPLEMENTATION

The digital implementation of both types of PLL structures is carried out using a universal controller for power electronics named Lin-X and shown in Fig. 8. It is a hardware platform based on a TI's TMS320F28335 32-bit digital signal processor (DSP).

Its design includes an isolated analogue front end for external voltage and current measurement (shown in Fig. 9), along with twelve isolated PWM outputs for the control of different types of power converters.

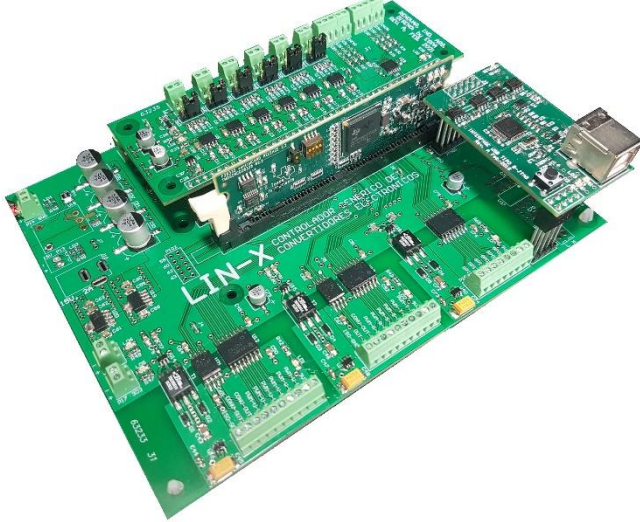


Fig. 8. Lin-X universal power converter controller.

Three voltage channels are used for the measurement of the grid voltage during the evaluation of the proposed PLL structures. The universal controller has also additional measurement channels for the control of grid-connected converters, such as bus voltage and phase currents.

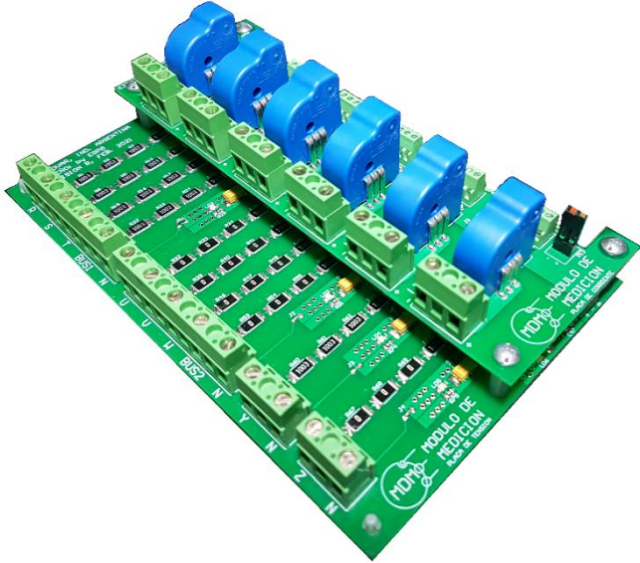


Fig. 9. Voltage and current measurement board.

V. SOFTWARE IMPLEMENTATION

The two PLL structures are translated to C code and programmed into the DSP of the controller by means of an integrated development environment (IDE) called Code Composer Studio.

A self-made library called *renolib* is used for this purpose. It contains a digital implementation of the most common control operations in power electronics: Park and Clarke transforms, integrators, low- and high-pass filters, proportional and integral controllers, limiters, etc. The designer must interconnect the different blocks according to

its control law, without the need of dealing with its digital implementation.

The use of this library allows rapid prototyping of control algorithms in power converters while using checked, debugged and version-controlled functions. A repository is used for documenting all library changes, updates and bug fixes.

An example of a *renolib* block is shown in Fig. 10. It consists of a definition for the digital implementation of a numerical integrator.

```
typedef struct{
    float in;           // Input
    float x[2];         // State variable
    float dT;           // Sampling time
}INTEGRATOR;
```

Fig. 10. Definition of structure INTEGRATOR in *renolib*.

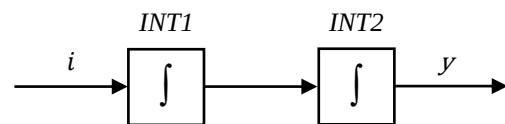
A C-macro is used for the execution of the resulting difference equation of the integrator, which depends on the discretization method used. Fig. 11 shows the resulting macro for the execution of the integrator by using forward Euler as discretization method.

```
#define INTEGRATOR_MACRO(v) \
    v.x[0] = v.in*v.dT + v.x[1]; \
    v.x[1] = v.x[0];
```

Fig. 11. Digital implementation of a numerical integrator in *renolib*.

Equations of different control blocks are periodically executed by means of an interrupt service request (ISR) in the DSP. Its period time must be equal to the discretization time of all dynamic blocks.

When the ISR is executed, a control system law is sequentially run. Designer must connect the different input and output ports of its blocks according to the proposed control system and execute them by calling their macros. Fig. 12 shows a brief example of the digital implementation for two cascaded integrators, arbitrarily named as *INT1* and *INT2*.



```
interrupt void ISR(void) {
    ...
    INT1.in = i;
    INTEGRATOR_MACRO(INT1);
    INT2.in = INT1.x[0];
    INTEGRATOR_MACRO(INT2);
    y = INT2.x[0];
    ...
}
```

Fig. 12. An example of continuous control system and its digital implementation with *renolib* library.

VI. RESULTS

The resulting electrical workbench for the evaluation of both PLL structures is shown in Fig. 13. It includes:

- A LV electrical network feed-in, by means of a three-phase transformer and autotransformer.

- A Lin-X universal power converter controller and its measurement board [11].
- A three-phase voltage source converter (to be controlled by the controller in future work).

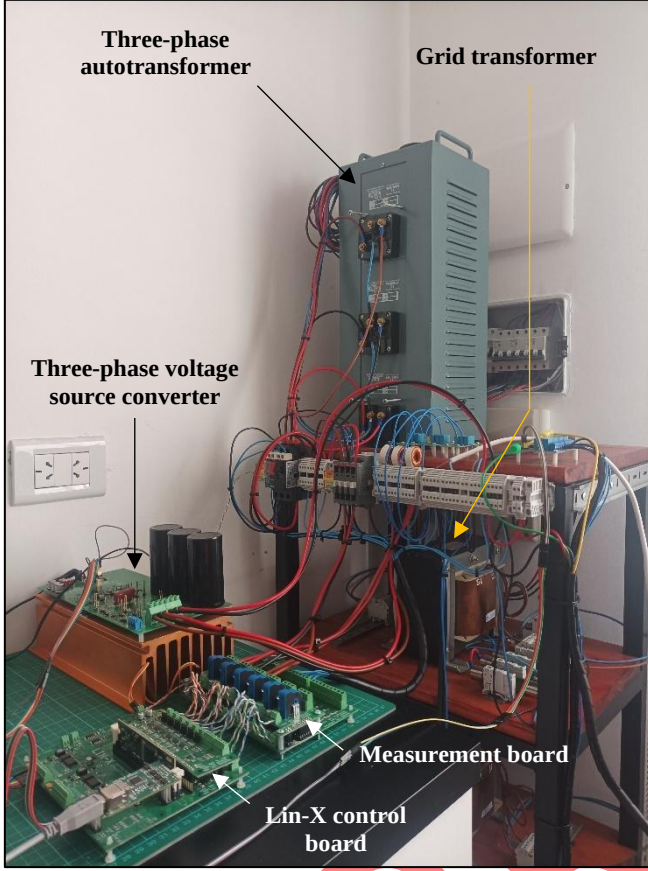


Fig. 13. Laboratory layout for the evaluation of PLL structures.

A. Digitally generated three-phase power system

Digital generation of a three-phase power system represents a starting point for the start-up of both proposed PLL structures. A symmetric and balanced three-phase power system is internally generated by the DSP and it serves as voltage inputs for the PLL structures under assessment.

$$\begin{cases} v_a = k_{sim} \cdot \sin(2\pi f_{sim} + \varphi_{sim}) \\ v_b = k_{sim} \cdot \sin(2\pi f_{sim} + \varphi_{sim} - 2\pi/3) \\ v_c = k_{sim} \cdot \sin(2\pi f_{sim} + \varphi_{sim} + 2\pi/3) \end{cases}$$

Variables k_{sim} , f_{sim} and φ_{sim} (simulated ac grid amplitude, frequency and angle, respectively) can be modified at will by designer for different sort of tests.

Since both PLL structures are digitally implemented, a digital to analogue conversion (DAC) is needed to record its dynamic performance in an oscilloscope during the tests. The universal power converter controller provides up to six pulse width modulated digital-to-analogue converters (PWM-DAC) for this purpose. This process is shown in Fig. 14.

A steady state condition is firstly evaluated. Main estimated variables, such as voltage magnitude, frequency and phase, are monitored and compared to generated values. Both magnitude and frequency must be constant values, whereas the estimated phase is expected to be a sawtooth waveform between 0 and 2π .

Once the PLL structures are properly working in steady state, the second part of this test consists of its dynamic tuning. The purpose of tuning is to improve the dynamic response of the overall system while keeping it stable. The dynamic response of a PLL system must be fast enough to assure the phase-locked condition even in grid faults and major disturbances in electrical power systems.

A simulated phase jump in the grid voltage is used for testing the dynamic response of both PLL structures. A 15° step change φ_{sim} is applied for this purpose. The dynamic response of SRF-PLL is shown in Fig. 15, whereas for EPLLII is presented in Fig. 16. Both figures show the estimated variables for each PLL: magnitude (yellow trace), frequency (cyan trace) and phase error (purple trace). Phase error is the difference between the phase of the simulated ac system and the phase calculated by the PLL. This error variable will be set to zero in steady state by both PLL.

A comparison between the dynamic response of both PLL shows that dynamic response of the EPLLII is faster than SRF-PLL, setting the phase error to zero is approximately 15ms. Besides, instantaneous deviations in estimated amplitude and frequency are clearly reduced when compared to SRF-PLL.

B. External ac grid

The use of an external ac grid enables the evaluation of both type of PLL and its response to unbalanced and distorted electrical grids.

The external ac grid is fed from the low voltage distribution network (3x380V / 50Hz, for residential use) as shown in Fig. 17. An isolation transformer is used for avoiding electrical hazards since the workbench is expected to be used with grid-connected converters. A three-phase

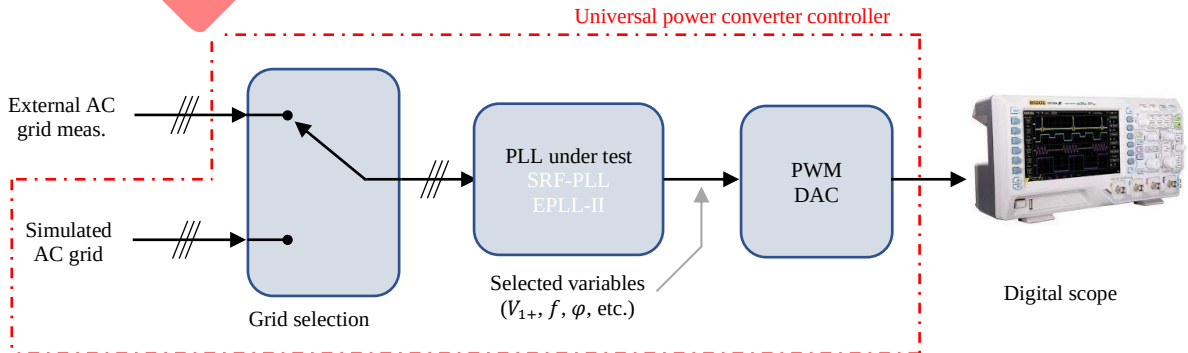


Fig. 14. Grid selection and debug process for the proposed PLL.

autotransformer allows the user to increase and decrease the magnitude of the measured voltage.



Fig. 15. Dynamic response of SRF-PLL to 15° phase jump in grid voltage.

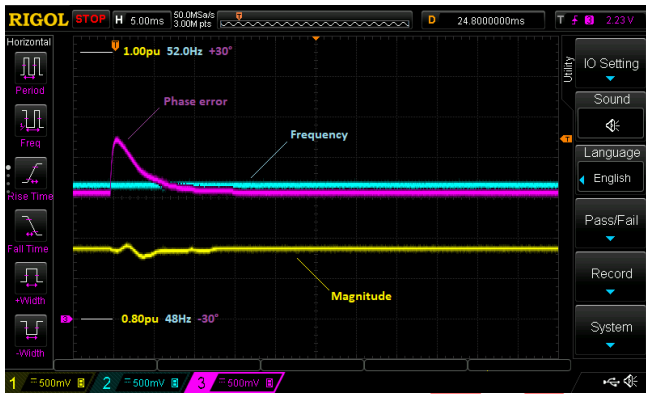


Fig. 16. Dynamic response of EPLLII to 15° phase jump in grid voltage.

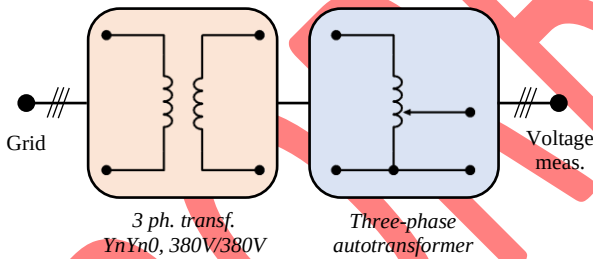


Fig. 17. External electrical grid.

Prior to the implementation of both PLL structures, the existing power quality of the external ac grid is evaluated by means of a grid analyser. Fig. 18 plots the three voltage waveforms whereas Fig. 19 shows its relative harmonic spectre.

As expected, the most significant odd harmonics are third, fifth, seventh and ninth, as all of them are larger than a 0,8%. All other harmonics are below 0,4% and can be considered as negligible.

The steady state response of the SRF-PLL (magnitude, frequency and phase) while using the external ac grid is shown in Fig. 20. It shows a ripple component in the estimated magnitude and frequency of grid voltage. Ripple frequency is 300Hz (six times the electrical frequency of the grid) and its magnitude is approximately 3% of the fundamental component. On the other hand, phase ripple is largely attenuated by means of the phase integrator located in the SRF-PLL, acting like a low-pass filter for this frequency component.

The presence of these ripple components in the estimated variables will negatively affect the implementation of grid-connected power converter control strategies, such as voltage-oriented control (VOC).

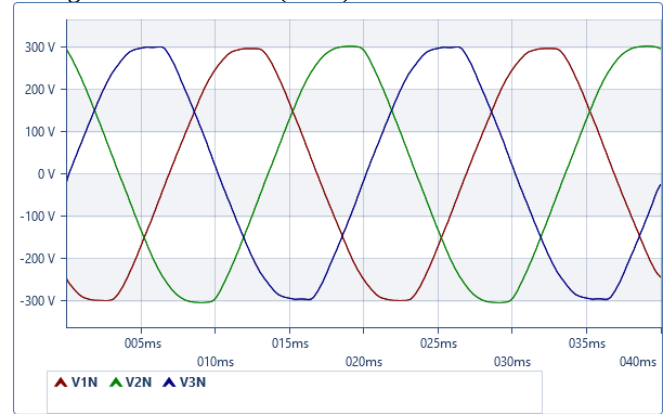


Fig. 18. Three-phase power system waveforms for evaluation of proposed PLL under steady state.

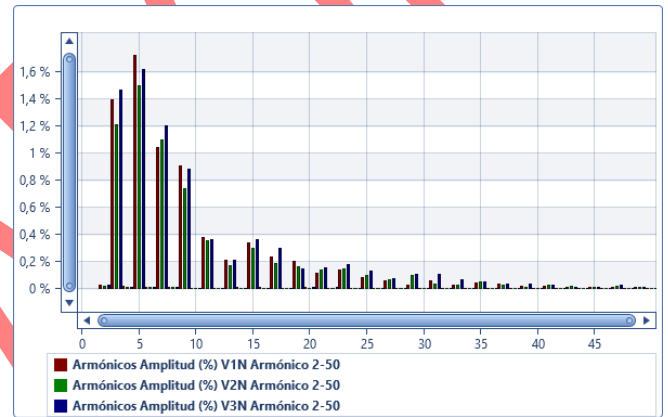


Fig. 19. Average harmonic spectre for the three-phase power system used.

The ripple is caused by the simultaneous appearance of two different harmonics on the measured grid voltage: 5th and 7th. The first of them corresponds to a negative sequence whereas the latter is a positive sequence. The grid analyzer used during the test is unable to determine the type of sequence for each harmonic. This sort of sequence identification is carried out by trial and error while defining the cores to be included in the implementation of EPLLII.

The digital implementation of EPLLII considers four cores for different sequences and harmonics, as listed below:

- Fundamental positive sequence, V_{1+} .
- Fundamental negative sequence, V_{1-} .
- Fifth harmonic negative sequence, V_{5-} .
- Seventh harmonic positive sequence, V_{7+} .
- DC removal.

The EPLLII allows the introduction of a special core for removing the dc component in the measured voltage. This component is mainly generated by the analogue front end (voltage offset) of the measurement board. The main advantage of this feature is that dc removal core does not introduce any sort of phase delay to the measured voltages.



Fig. 20. Steady state response of SRF-PLL while connected to the grid.

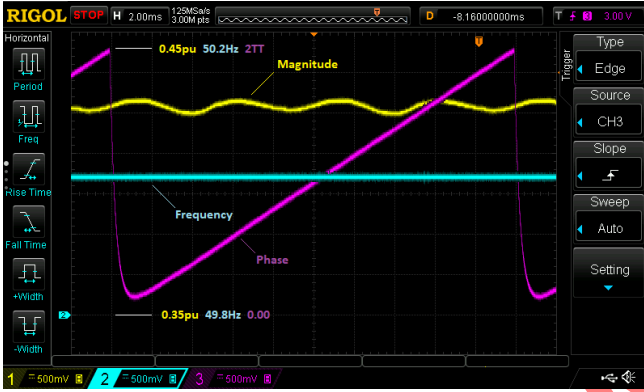


Fig. 21. Steady state response of EPLLII while connected to the grid.

The selection of the cores is based on the determination of the most relevant harmonics shown in Fig. 19. The increasing number of cores clearly extends the computing time for the DSP. As the computing time extends, there may be the necessity to reduce ISR frequency, affecting in a negative way the process of discretization. Table 1 compares the execution time of both PLL configurations assuming a 10kHz ISR frequency.

PLL configuration	ISR
SRF-PLL	36 %
EPLLII	58%

Table 1. ISR execution time for proposed PLL structures.

The results for EPLLII clearly show a reduction in the magnitude of the pulsating components in both magnitude and frequency estimation of fundamental positive sequence of the grid voltage.

The estimation of voltage harmonics by the EPLLII structure allows the converter controller to provide more complex functionalities in grid-tied converters such as current harmonic suppression and active filtering.

VII. CONCLUSION

Agregar más contenido (introducción).

The availability of multiple blocks in *renolib* library allows the rapid prototyping of control systems in a universal power converter controller. The library is intended for its use in several applications of power electronics such as: variable speed drives, grid-connected inverters, switching mode power supplies, uninterrupted power supplies, industrial rectifiers and so on.

The library is used for the digital implementation of two phase-locked loop systems: SRF-PLL and EPLLII. Results has shown that the presence of harmonics and voltage unbalance may affect in a negative way the grid synchronization process in simpler PLL structures such as the SRF-PLL.

More complex structures, like EPLLII, allows the estimation of multiple sequences and harmonics, enabling for its accountancy and calculation and, therefore, its cancelation in grid synchronization process. As a result, the estimated grid variables show a reduced ripple compared to simpler PLL structures.

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